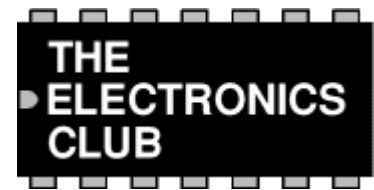


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Frequently Asked Questions



Why is the letter I used to represent current?

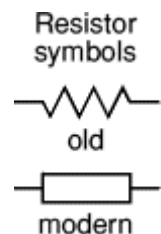
The letter I seems to be an odd choice for the English language, but it was chosen in the early days of electricity to represent **intensity of current** which we simply call current today. The unit of current, the ampere, is named after the French scientist André-Marie Ampère in recognition of his work on the relationship between electric current and magnetism. Ampère referred to electric current as "**l'intensité du courant électrique**", so I was a logical choice to represent intensité (intensity). I am grateful to Barry Caruth for suggesting a search of the internet for "Ampère" and "l'intensité du courant électrique" which returns many sites as evidence (most of them French) enabling me to answer this question with confidence.

Further information: [Quantities used in Electronics](#) | [Voltage and Current](#) | [Ohm's Law](#)

Why do some books use zig-zag lines for resistors in circuit diagrams?

The zig-zag line is the old symbol for a resistor and you may find it in older books and magazines. Unfortunately a few publications still use it! The correct modern symbol for a resistor is a rectangle.

Further information: [Resistors](#) | [Circuit Symbols](#) | [Circuit Diagrams](#)



My project has a resistor labelled 47, does that mean 47kΩ?

No, it means 47Ω which is 1000 times smaller! 47kΩ would be shortened to 47k (or 47K). The ohm (Ω) symbol is often omitted from circuit diagrams and component layouts but the k (meaning kilo = 1000) will always be included if it is needed.

Further information: [Resistors](#)

Why do resistors have odd values like 47k and 56k, but not 50k?

There is a good reason for these odd values and it is explained on the [Resistors](#) page.

A project on another website lists a 10kW resistor! What does it mean?

It almost certainly means a 10kΩ resistor. This is a common error which occurs when the web page specifies a Greek font. If this font is not available on your computer you see the character in your standard font and it happens to be W which is the symbol for watt, the unit of power! I avoid the problem on this website by using a small image for Ω. In a few projects a low value resistor with a high power rating is required but the power will be something smaller like 5W, never 10kW which is more powerful than an electric heater!

Further information: [Resistors](#)

My soldering iron was supplied with a hook, do I really need to buy a stand as well?

For safety you must buy (or make) a stand for your soldering iron. Please don't use the hook because it leaves exposed the very hot element and tip of the iron - it is too easy to accidentally touch them and burn yourself. If you can't afford to buy a stand you could try making your own with a spiral of stiff galvanised iron wire (a coat-hanger?) screwed to a block of wood. Ideally the stand should include a damp sponge for safely wiping the tip of the iron when it needs cleaning.



Soldering iron stand
Photograph © [Rapid Electronics](#).

Further information: [Tools for Electronics](#) | [Soldering Guide](#)

Where can I buy heatproof cable to replace the ordinary cable on my soldering iron?

Silicone heat resistant cable is sold in 1.5 metre lengths for exactly this purpose by [Rapid Electronics](#), part number 85-0590 (look in the Soldering Equipment section). If you use another supplier make sure you buy 3-core mains flex with a current rating of 3A (the proper name for mains appliance leads is flex, not cable). Please note that to change over to the new flex you will need to borrow a second soldering iron! This is because the flex is soldered to the iron's element. Make sure that you connect the wires correctly in the iron and in the mains plug which should have a 3A fuse.

Further information: [Tools for Electronics](#) | [Soldering Guide](#)

My teacher says that Christmas tree lights are a series circuit, so when one lamp blew on Christmas Eve why didn't they all go out?

Traditional Christmas tree lights are connected in series and you are correct in thinking that if one lamp blows all the lamps should go out. The problem is that Christmas tree lights are not like ordinary lamps! When they blow they automatically short-circuit (they become like a wire link) so the circuit is still complete and the other lamps remain lit. This makes it easy to see the blown lamp, but do remember to switch off before changing it.

Teachers - please be careful when giving Christmas tree lights as an example of a series circuit!

Further information: [Lamps](#) | [Series and Parallel Connections](#)



What component has a black stripe in the centre (it looks like a diode)?

A small component about the size of a resistor or signal diode with a single black stripe in the centre is a **zero-ohm resistor**, it is really just a wire link! These components are used on commercial PCBs because they are easier for machines to handle than small pieces of wire. The single black stripe is logical because it means zero in the resistor colour code. Ordinary resistors have at least four stripes. Diodes have a single stripe near one end, not in the centre.

Further information: [Diodes](#) | [Resistors](#)

What is a "short circuit"?

A "short circuit" is a connection of very low resistance such as a wire (almost 0Ω) which

provides a very easy path for current. Think of it as an electrical short-cut. It is normally used to describe a fault or accidental connection rather than a deliberate one.

For example: if the leads from a battery touch one another they create a very low resistance connection across the battery, so we say they have caused a short circuit across the battery. Current will flow through this short circuit rather than through the proper circuit. This stops the circuit working and it may cause a fire because the leads and battery will become hot with a large current flowing.

Further information: [Voltage and Current](#) | [Resistance](#)

What does "open circuit" mean?

"Open circuit" means no connection. It is usually used to describe a break in some part of a circuit which could be deliberate (such as a switch in the open or off position) or a fault (such as a broken wire or burnt out component).

Further information: [Voltage and Current](#) | [Resistance](#)

How do I choose a relay to use with one of your projects?

The 555 timer chip used in many projects can supply current up to 200mA so it can power most relays directly. However, you must connect a signal diode (a 1N4148 for example) in parallel across the relay coil to protect the 555. Note that this diode is connected 'backwards' so that it will normally **not** conduct.

Further information: [Relays](#) | [Diodes](#) | [Model Railway Signal](#) (example project with relay)

I want to use a large number of LEDs, do I need a resistor for each one?

No, you can usually connect a few LEDs of the same type in series and just use one resistor. The number of LEDs you can connect in series depends on the circuit's supply voltage. This arrangement has the advantage of reducing the total current required by the circuit. Please see the LEDs page for more details:

Further information: [Connecting LEDs in series](#)

I want to build up a stock of components, what should I buy first?

Most people build their first few projects from complete kits, but if you want to try adapting published projects or designing and building your own circuits you will need to have a small stock of components available. There is a page with advice on buying a [starter kit](#) of components.

Further information: [Starter kit](#) | [Links](#) (for suppliers)

What is a Darlington pair?

A Darlington pair is two transistors connected together so that the current amplified by the first is further amplified by the second transistor, giving a very high gain of 10000 or so.

Further information: [Transistors](#)

What does 'sinking a current' mean?

It means current is flowing into the output of a chip. This happens when the output is low (0V) if there is a device connected between the positive supply (+Vs) and the output. It is the opposite of sourcing a current which means current is flowing out of the output. Most chip outputs can both sink and source current.

Further information: [Integrated Circuits \(Chips\)](#)

Are 'time period' and 'time constant' the same thing?

No, they have different meanings although both are time. **Time period** is the duration of a single pulse or the time for one cycle of a repeating electrical signal. **Time constant** is a property of a changing system, such as a capacitor charging and discharging.

Further information, **time period**: [Electrical Signals](#) | [Astable](#) | [Monostable](#) | **time constant**: [Capacitance](#)

What is a PIC?

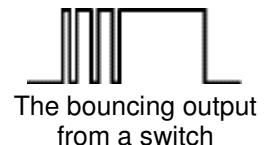
A PIC is a **P**rogrammable **I**ntegrated **C**ircuit microcontroller, a 'computer-on-a-chip'. They have a processor and memory to run a program responding to inputs and controlling outputs, so they can easily achieve complex functions which would require several conventional ICs. I can strongly recommend the PICAXE system because it is easy to program (and re-program) the PICs with a standard computer - no specialist equipment is required other than a low-cost download lead. The programming software and extensive documentation is available to download free of charge, making the system ideal for education and users at home.

Further information (including downloads): www.picaxe.co.uk

Why does my circuit count 3 or 4 when I press the switch once?

This is likely to happen if a switch is connected directly to the clock input of a counter. Switches contacts tend to rapidly bounce open and closed a few times when the switch is operated. The counter sees this as several clock pulses, not the single pulse you expect. One solution is to make the switch trigger a monostable circuit with a short time constant (0.1s for example) and use this to drive the clock input.

Further information: [Counting Circuits](#) | [555 Monostable Circuit](#)



What software do you use to draw the circuit diagrams?

I created the circuit diagrams (and all the other drawings on the website) using a vector drawing application called Draw. It comes as standard with all [RISC OS](#) computers, but is not available for Windows computers. The diagrams are converted to GIFs for the website for maximum compatibility.

Further information: [Circuit Diagrams](#) | [RISC OS](#)

What is a schematic?

It is a circuit diagram.

Further information: [Circuit Diagrams](#)

What does 'SMD' mean?

'SMD' means Surface Mount Device. SMDs are components with small pads instead of leads for their contacts. They are designed for soldering by machine onto specially designed PCBs and are not suitable for educational or hobby circuits constructed on breadboard or stripboard. Do not buy SMD components for your projects!

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